Regular Language / finite Automata

FA → With O/P → Moore (each state has o/p), Mealy → Both are deterministic.
→ Without o/p → NFA
→ DFA
→ ∈-NFA

| $\|w\| \bmod n = 0$.
states = n
++

Ques Set of string over {0,1,2} & divisible by 4.

	0	1	2
q_0	q_0	q_1	q_2
q_1	q_3	q_0	q_1
q_2	q_2	q_3	q_0
q_3	q_1	q_2	q_3

Imp:
→ Complements works only on DFA. not for NFA.

→ If L_1 = NFA then $(L_1)^R$ → The resultant can be NFA or DFA.
doubt
↓ Not true.

# states	# states
n	2^n [max^m]
NFA	DFA
	↓
	[n to 2^n]

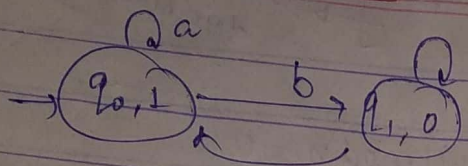
Imp ++
→ If Ask for minimal no of states in FA without mentioning that NFA or DFA. → Go with NFA.

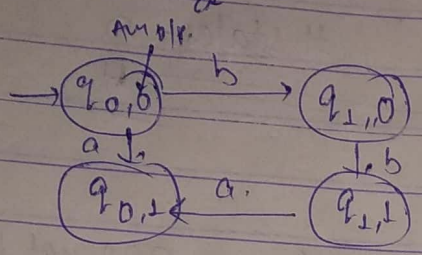
Imp:
→ Complementing NFA may leads to violation of Language.
NFA − m − L_1 &
NFA − $\overline{m}$ − L_2 ⇒ then $L_1 = L_2$ or $L_1 \neq L_2$ → can't decide.

Minimization of DFA:
→ delete all un-reachable state.
→ find out Equivalence classes
[Non-final, final]

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Moore Machine



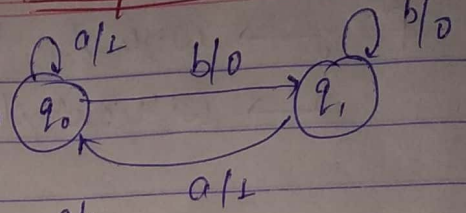


↳ Input length = output length

n n+1

Mealy Machine

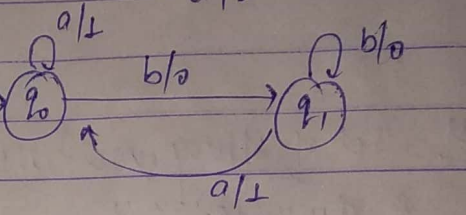
States doesn't change.



States must increased

$(N \times m) \rightarrow$ Max^m states

$\#I/Ps$ ++



↳ Input length = o/p length

∈-NFA to NFA :

$\in^*$	0	$\in^*$
A → A	→ A	→ A, B
→ B	→ C	→ C, D
→ D	→ D	→ D.

⇒ So, for A & 0. (A, 0) ⇒ (A, B, C, D)

~~Ex~~ → All states which you can reach by only ∈. All those will be final state.

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Regular Lang / Grammer → Type 3 Grammer

- Right linear Grammer
- LLG.

Rules

- $\phi + R = R + \phi = R$
- $\phi \cdot R = \phi = R \cdot \phi$
- $\epsilon \cdot R = \epsilon \cdot R = R \cdot \epsilon = R$
- $\epsilon^* = \epsilon$
- $\epsilon + RR^* = R^*R + \epsilon = R^*$
- $\phi^* = \epsilon$
- $(a+b)^* = (a^*+b^*)^* = (a^*b^*)^*$
  $= (a^*+b)^* = (a+b^*)^*$
  $= a^*(ba^*)^* = b^*(ab^*)^*$

Lup:

- FA → RLG. (L) and L^R → that is $\bar{L}$
- FA (L) → RLG (L) —Reverse→ LLG (L^R)
- FA (L) —R→ FA (L^R) → RLG (L^R) → LLG $(L^R)^R$

Chomsky ⓐ

- $A \to \alpha B | \beta$ or $A \to B\alpha | \beta$

  ⇓

  Either fully RLG or LLG not Combination.

Lup

- Every finite lang. is Regular.
- Pumping lemma → for infinite lang. (should follow a particular pattern).

Pumping lemma → Yes — May or may not be Regular.
→ No → if not present any pattern, then it ~~may be regular or may not be~~ will not be regular.

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Imp. Regular Lang:

Que. $(111+11111)^*$ → # states in DFA?

✓ (1) $ww^R \mid |w|=2$

✗ (2) $ww^R \mid w \in (a,b)^*$

✗ (3) $ww \mid w \in (a,b)^*$

✗ (4) $w \mid n_a(w) = n_b(w)$ (Imp: $\leq$, $\geq$) → Unbounded Counting

✗ (8) $www^R \mid w \in (0,b)^*$

✓ (7) $wxw^R \mid w,x \in (0,1)^+$

✗ (8) $wxw^R \mid w \in (0,1)^+ \& |x|=2$

✓ (9) $xwwy \mid x,y,w \in (0,1)^*$

✗ (10) $xww^Ry \mid w \in (0,1)^+, |x|=2, |y|=3$

✗ (11) $ww^Rx \mid w,x \in (0,1)^+$

✗ (12) $xww^R \mid x,w \in (0,1)^+$.

✓ (5) Imp: $n_a(w) \bmod 3 \leq n_b(w) \bmod 3$

↳ Modulus Counting.

Imp Que

Sets ~~Not~~ Countable? $\Sigma = \{0,b\}$

✓ → Set of all strings over Σ. (Countable & finite)

✗ → Set of all lang" (Uncountable)

✓ → Set of all Regular lang.

✓ → Set of all lang. over Σ accepted by turing machine.

Imp. About Regular Lang.

→ Every lang. has a regular superset. $[\Sigma^*]$

→ Every Lang. has a Regular subset. $[\phi]$

→ Every subset of a regular lang may not be Reg. $[a^nb^n \leq \Sigma^*]$

→ Every subset of a finite lang is Regular. [Also finite].

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Pushdown Automata: CFL & DCFL → Type 2.

→ Remove Unit prod^n
→ ϵ
→ Useless Symbol.

→ PDA with limited stack size is Regular. [DFA].

→ $A \to \alpha$, $\alpha \in (V \cup T)^*$ ⇕ CNF.
  ↳ eg. $A \to BC$, $A \to a$.

, $A \to a\,(V \cup T)^k$ ⇓ GNF.

CFG
steps req. $2|W|-1$.
steps req. $|W|$
CNF (Chomsky)
GNF (Griebach)

CNF: $A \to BC$, $A \to a$ ⇓ Length Restricted to 2.

GNF: $A \to a\alpha$, $\alpha \to V^*$.
eg. $A \to aBCDE$, $A \to a$.

Imp.

Note: WW is Context Sensitive Language.

Imp.
→ Check Notes (Detailed for Ex. of CFG).

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Que. $(111+11111)^*$ → # state :- -

↳ LBA.

Context Sensitive Lay → Type 1. ↳ Not Null.

$x \to y$, $y, x \in (V \cup T)^+$ and $|x| \leq |y|$.

Imp. : Noone find out DLBA & NLBA are equivalent or not ##

Lay. $\{a^n b^n c^n, n \geq 1\}$

$\{a^{n!} ; n > 0\}$

$\{a^n ; n = m^2\}$

$\{a^n ; n$ is prime$\}$

$\{ww ; w \in (a,b)^+\}$

$\{w^n ; w \in (a,b)^+\}$

$\{wwwR ; w \in (a,b)^+\}$. (##)

Imp.

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Turing Machine, RE, REC

- doesn't Accept ϵ.
- Type 0.
- $x \to y$; $y \in (VUT)^*$ & $x \in (VUT)^+$ ++
- Un-Restricted.

→ FA + 2 stacks : TM
→ FA + 1 stacks : PDA ⟶ (No limited size of stack).

→ TM + Tape (Part in which I/P is present) ⟹ Linear Bound Automata.

→ If L is RE but not R, then $\bar{L}$ not RE. → Ip

Ip : → If a Lay. L and its Compliment $\bar{L}$ are recursive enumerable, then both Lay. are recursive.

→ If L is recursive, then $\bar{L}$ is also recursive and Consequently both are recursive enumerable.

→ No membership Algo for RE but for Recursive it exist.

Rice Theorem: Any non-trivial property of L(TM) is Undecidable.

A non-trivial property is satisfied by some RE and not satisfied by some others.

Ip (on tutorial point).

Property P is set of languages.

A non trivial property, where L∈P, both the following property should hold-

1. There exist TM, M1 & M2 that recognize same Lay. either ($\langle M1 \rangle, \langle M2 \rangle \in L$) or ($\langle M1 \rangle \langle M2 \rangle \notin L$)
2. There exist TM, M1 & M2 where M1 recognize the lay while M2 doesn't.

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→ Every TM is finite & can be represented using (0,1).

Note: L_3 : $\langle M \rangle$ | M accept ε.

fip → M is FM that accepts a string of length 2014.

→ These are non-Trivial

So, Undecidable.

fip** P ⊆ NP ⊆ Recursive

So, NP are decidable.

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Countability

→ Every set of Countable set is either finite or Countable.

Dfn → Every Lay is Countable. $L \subseteq \Sigma^*$.

→ Set of all Lang is not Countable. i.e. 2^{Σ^*}.

→ Every string $\Sigma^* = \{0,1\}$ are Countable.

→ Set of all TM are Countable

⇒ REL, RL, CSL, CFL, RL → are Countable.

⇒ TM, LBA, PDA, FA

Inf theorem: If 's' is Countable infinite then 2^S is Uncountable.

Σ^* (Countable Infinite) 2^{Σ^*} (Uncountable)

→ If S_1 and S_2 are Countable then $S_1 \cup S_2$, $S_1 \times S_2$ are Countable.

→ The Cartesian product of finite no of Countable set is Countable.

→ The set of all Lay. which are not RE are Uncountable.

◉ ज्ञ... ...पूर्ण प्रश्न पत्रों का समावेश ◉ सरल भाषा में ...
◉ अनुभवी अनुदेशक गण द्वारा उत्तरों का स्पष्टीकरण ◉ प्रशिक्षुओं के कौशल में वृद्धीकारक
◉ प्रथम बार उत्तरों का स्पष्टीकरण सहित सॉल्वड पेपर ◉ सरल भाषा मे लिखित

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Decidability

→ Given any instance of a problem, it is always decidable.
[But if domain is Given (set of all Effective), then it may decidable or not.].

Sup.

$P_1 \xrightarrow{\Delta} P_2$

→ if P_2 is RE/R/Decidable/NP/P/Hard then P_1 also same.

→ if P_1 is not (_____) then P_2 is also not.

→ only these we can decide other conditions we can't say anything. ++

Some un-decidable problem:

→ Halting problem is un-decidable.

→ State entry problem.

↓

Partially decidable [RE but not Recursive]

Partial Decidable:

↓

If halt for some input.

Sup

→ Every Recursive Lang is Computable but reverse not true.

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CYK Algo for membership

Problem	Reg	DCFL	CFL	CSL	Recursive	RE
→ Does $w \in L$? (Membership)	D	D	D	D	D	UD
→ Is $L = \phi$ (Emptiness)	D	D	D	UD	UD	UD
→ Is $L = \Sigma^*$ (Completeness)	D	UD	UD	UD	UD	UD
→ Is $L_1 = L_2$ (Equality)	D	UD	UD	UD	UD	UD
→ Is $L_1 \subseteq L_2$ (Subset)	D	UD	UD	UD	UD	UD
→ $L_1 \cap L_2 = \phi$	D	UD	UD	UD	UD	UD
→ Is L is finite or not?	D	D	D	UD	UD	UD
→ Is Compliment of 'L' is Lay of same type or not.	D	D	UD	D	D	UD
→ Is intersection of two Lay is same type or not.	D	UD	UD	UD	UD	UD
→ Is L regular Lay?	D	D	UD	UD	UD.	UD

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Homomorphism: $h(h^{-1}(L)) \subseteq L$.

Closure Properties

Property	Regular	CFL	DCFL	CSL	R	RE
$L_1 \cup L_2$	✓	✓	X	✓	✓	✓
$L_1 \cap L_2$	✓	X	X	✓	✓	X
$L_1 - L_2$	✓	X	X	✓	✓	X
$\overline{L}$	✓	X	✓	✓	✓	X
Intr^n with Regular	✓	✓	✓	✓	✓	✓
$L_1 \cdot L_2$	✓	✓	X (X)	✓	✓	✓
L_1^*	✓	✓	X	✓	✓	✓
L_1^+	✓	✓	X	✓	✓	✓
L^R (Reverse)	✓	✓	X	✓	✓	✓
Homomorphism	✓	✓	X	X	X	✓
ϵ-free Homo	✓	✓	X	✓	✓	✓
Substitution	✓	✓	X	X	X	✓
ϵ-free substitution	✓	✓	X	✓	✓	✓
Inverse Homo	✓	✓	✓	✓	✓	✓
Right Quotient	✓	⊙				

How to find this.

Imp: RL is not closed under infinite Union of RL. (+)

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